

Vijay Mohanram Iyer

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SUMMARY

Graduate engineer specializing in intelligent safety systems and health applications powered by computer vision and machine learning. Skilled in transforming advanced algorithms including deep learning architectures and large language models into practical, user-centered tools that enhance security and promote human well-being.

PROFESSIONAL EXPERIENCE

1. FZI *September 2025 – November 2025* **Research Assistant**

- Implement dynamic ROI tracking using 3D geometric facemesh reconstruction and facial landmark localization to improve spatial awareness in DeepFake detection.
- Design an interactive visualization interface that maps artifact energy across facial regions using image-based feature analysis and temporal coherence cues.
- Optimize detection models through transfer learning, fine-tuning and ensemble methods to enhance performance on challenging visual manipulations and research publication.

2. TecoLab *March 2023 – September 2025* **Working Student**

- ML4Print: Classification based on printer-specific characteristics and paper substrate type using feature extraction and supervised ML classifiers.
- Heat simulation: utilized regression models and CFD data integration to simulate the thermal behavior of liquids within industrial valves, thereby enhancing predictive maintenance capabilities and system reliability.
- Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.

3. Access@KIT *March 2023 – May 2023* **Working Student**

- Designed and implemented a bi-directional LSTM network for text-to-speech synthesis, optimizing acoustic feature prediction to capture temporal dependencies and prosody for natural, intelligible speech.
- Performed data preprocessing including phoneme alignment and feature extraction, and evaluated model performance with iterative tuning to enhance digital accessibility for visually impaired users.

4. Accenture India Pvt. Ltd. *February 2022 – April 2022* **Associate Software Engineer**

- Maintained IBM Mainframe systems with batch job scheduling and performance monitoring scripts.

5. Accur Digitus
Web Developer Intern

January 2020 – May 2020

- Developed React.js web applications with component-based architecture and state management.
- Implemented RESTful APIs using JSON and asynchronous fetch for front-end/back-end integration.

EDUCATION

M.Sc. in Electrical Engineering and Information Technology
Karlsruhe Institute of Technology
GPA: 2.3

May 2022 – July 2025

Bachelor of Engineering in Electronics Engineering
University of Mumbai, India
GPA: 2.8

August 2017 – May 2021

PROJECTS

1. **Attention Mechanism and Multi-ROI Artifact Recognition for DeepFake Detection Using rPPG (FZI)** *February 2025 – July 2025*
This Master Thesis aimed to develop an end-to-end deep learning pipeline using rPPG signals for artifact evaluation. Proposed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification. Benchmarked under real-world conditions.
2. **CamCussion: Eye Tracking Software (Zeiss Innovation Hub)** *November 2023 – July 2024*
Utilized computer-vision to analyze pupil dilation and saccadic eye movements in real time to track and assess eye behavior, contributing to accurate concussion diagnosis.
3. **Self-Driving Car using LIDAR** *September 2022 – February 2023*
Developed a real-time 3D obstacle detection using LIDAR data in autonomous driving. It transforms raw point clouds into ViT-compatible tokens via efficient voxelization and applies Multi-head attention to capture global context, improving detection accuracy.
4. **Real-Time Car Accident Alert System** *January 2022 – June 2022*
Developed an embedded LSTM-based accident detection system running on ESP32 using TensorFlow Lite for Microcontrollers, enabling real-time crash detection and automated SMS/email alerts with precise GPS location.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, React.js, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, ROS2
- **ADAS:** Sensor fusion, camera and LiDAR calibration, perception algorithms, path planning, object detection & tracking
- **Tools:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX
- **Spoken Languages:** English C1, German A2 (Learning)

SOFT SKILLS

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging, Balancing and Prioritizing tasks to meet goals.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.